§1.4. 이항계수 binomial coefficients HW. 9, 11, 16

Def. $\alpha \in \mathbb{R}$, $k \in \mathbb{Z}^{\geq 0}$

$$\binom{\alpha}{k} := \frac{\alpha(\alpha-1)\cdots(\alpha-k+1)}{k!} =: \alpha^{\underline{k}}$$

e.g. $n \in \mathbb{Z}$. $\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{n(n-1)\cdots(n-k+1)}{k(k-1)\cdots 1}$, $\binom{n}{0} = 0$

Thm 1.20. Let S be a set containing exactly n elements. For any $k \geq 0$, the number of subsets of S containing precisely k elements is $\binom{n}{k}$.

pf) # of permutations of S $= n(n-1)\cdots 1 = n!$

Let's count this in other way.

i) Choose a subset of $(n-k)$ elements in S : X ways

ii) Arrange its element in various orders : $(n-k)!$

iii) " the other elements " : $k!$

$\therefore$ $n! = X\,(n-k)!\,k! \Rightarrow X = \frac{n!}{k!\,(n-k)!}$ $\square$

Thm 1.21 The product of any k consecutive integers is divisible by $k!$.

pf) $\overset{0 \leq k < n}{n(n-1)\cdots(n-k+1)} = \binom{n}{k} k! \ (\in \mathbb{Z})$, $\binom{n}{k} \in \mathbb{Z}$

$\therefore$ $k!$ 의 배수.

what if $0 \leq n < k$ or $n < 0$. ///

Thm 1.22 (Binomial Thm) $n \geq 1$. $n \in \mathbb{Z}$

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$$ 이항정리

pf) $(x+y)^n = (x+y)(x+y)\cdots(x+y) \rightsquigarrow \bigcirc\, x^k y^{n-k}$

choose k terms out of n terms $= \binom{n}{k}$ $\square$